Semantic Segmentation in Pontryagin Spaces

Random Feature Embeddings with Indefinite Metric for UAV Crop Imagery

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1 Introduction

Deep neural network classifiers typically embed intermediate representations in Euclidean space and apply a linear decision rule. A growing body of work has shown that certain data distributions — in particular those with latent hierarchical or hyperbolic structure — are better captured by non-Euclidean geometries. Hyperbolic spaces (Poincaré ball, hyperboloid) have been explored for this purpose [3, 4], but they impose a *constant negative* curvature that may not suit all problems.

This work proposes an alternative: embedding pixel features into a **Pontryagin space** Π_κ, a flat pseudo-Riemannian manifold equipped with an *indefinite* inner product of signature $(\cdot, \kappa)$. Unlike the Poincaré ball, a Pontryagin space is both flat and able to represent mixed-curvature structure through the coexistence of positive- and negative-definite subspaces. The embedding is constructed from two families of random features:

- **Random Fourier Features (RFF)** [1] approximate a positive-definite RBF kernel K_+.
- **Spherical Random Features (SRF)** [2] approximate a polynomial kernel K_- of degree d_{poly} on the unit sphere.

The resulting feature vector lives in $\Pi_\kappa = \mathbb{R}^{p+\kappa}$ with metric $J = \text{diag}(+1^p, -1^\kappa)$, and classification is performed by *J-hyperplane* multinomial logistic regression.

The experimental testbed is binary semantic segmentation of sugarcane crops from UAV imagery, where all three heads (Euclidean, Hyperbolic, Pontryagin) share an identical deep FCN backbone and training protocol, isolating the effect of the embedding geometry. Interpretability of each model is assessed using **SegScoreCAM**, an adaptation of ScoreCAM for pixel-wise segmentation outputs.

2 Dataset

2.1 UAV Sugarcane Segmentation

The dataset consists of aerial images acquired by an unmanned aerial vehicle (UAV) over agricultural fields containing sugarcane, banana trees, pepper, spinach, and other crops. The annotation format is **YOLO polygon segmentation**: each label file contains class index and normalised polygon vertices per instance.

Preprocessing. All images are resized to 256×256 pixels using bilinear interpolation. Polygon masks are rasterised at the same resolution. To focus on a single, well-defined binary problem, the dataset is filtered to **binary sugarcane segmentation**: class 0 = background, class 1 = sugarcane; images without any sugarcane polygon are discarded. Pixel intensities are normalised with ImageNet statistics ($\mu = (0.485, 0.456, 0.406)$, $\sigma = (0.229, 0.224, 0.225)$).

Class imbalance. Background pixels are far more frequent than sugarcane pixels. A weighted cross-entropy loss compensates for this imbalance with weights $w_{\text{bg}} = 0.20$ and $w_{\text{sc}} = 1.00$.

Splits and augmentation. The dataset is split into train / validation / test partitions provided by the source repository, containing 81 / 30 / 20 images respectively (before binary filtering). After discarding images without sugarcane, the effective splits used in training are 49 train / 23 validation / 13 test images. During training, random horizontal and vertical flips (each with probability 0.5) are applied.

3 Backbone Architectures

Two backbone architectures are evaluated: a lightweight deep FCN for rapid ablation and a full U-Net for competitive benchmarking. All segmentation heads (Euclidean, Hyperbolic, Pontryagin) are compatible with both backbones; the U-Net experiments additionally include a Vanilla baseline that trains without class-weighted loss.

3.1 DeepFCN

All FCN segmentation heads share a **deep fully-convolutional network (DeepFCN)** backbone with configurable depth $D \in \{3, 4, 5\}$. Each stage consists of two 3×3 strided convolutions (stride 2) with batch normalisation and ReLU activation. Channel widths follow the schedule $3 \to 32 \to 64 \to 128 \to 256 \to 256$ (the last stage is repeated when $D = 5$). A final 1×1 projection maps the last stage to a fixed $C_{\text{out}} = 64$ channel dimension.

Given an input $\mathbf{x} \in \mathbb{R}^{3\times H\times W}$, the backbone produces a feature map $\mathbf{f} \in \mathbb{R}^{64\times H'\times W'}$ where $H' = H/2^D$. This map is then processed by the segmentation head, and the resulting low-resolution logits are bilinearly upsampled to the original resolution before computing the loss.

3.2 U-Net Backbone

The U-Net backbone (Ronneberger et al., 2015) is an encoder-decoder architecture that preserves full spatial resolution through skip connections, avoiding the upsampling artefacts common in FCN-style designs.

Encoder. D double-conv blocks (two 3×3 convolutions, each followed by batch normalisation and ReLU), each followed by 2×2 max-pooling. The channel schedule is $C_k = C_{\text{base}} \cdot \min(2^k, 8)$ for stage $k = 0, \ldots, D-1$, where $C_{\text{base}} = 64$ by default. For $D = 4$ this gives $64 \to 128 \to 256 \to 512$.

Bottleneck. A single double-conv block at the lowest resolution: $C_{D-1} \to C_D = C_{\text{base}} \cdot \min(2^D, 8)$ (e.g. $512 \to 1024$ for $D = 4$).

Decoder. D upsampling stages, each performing bilinear upsampling followed by skip-connection concatenation with the matching encoder output, and then a double-conv block. The output is projected to $C_{\text{out}} = 64$ channels by a 1×1 convolution.

The backbone produces $\mathbf{f} \in \mathbb{R}^{64\times H\times W}$ — at the *same* spatial resolution as the input, so no further upsampling is needed. The depth $D \in \{3, 4, 5\}$ is a hyperparameter optimised in the HPO phase.

4 Segmentation Heads

The same four heads are used with both backbones. The Vanilla head is a pure CE baseline without class weighting; it is included in the U-Net experiments to provide an upper bound untouched by any geometric inductive bias.

4.1 Vanilla (U-Net baseline only)

The Vanilla head applies a 1×1 convolution to the backbone features and minimises the standard (unweighted) cross-entropy loss:

$$\mathcal{L}_{\text{vanilla}} = -\frac{1}{N} \sum_{n=1}^{N} \log \frac{e^{(\mathbf{W}_{\text{cls}} \mathbf{f}_n)_{y_n}}}{\sum_c e^{(\mathbf{W}_{\text{cls}} \mathbf{f}_n)_c}}.$$

No class-weight correction is applied. This model is the strongest "geometry-free" reference: any improvement from the geometric heads must overcome a baseline that is otherwise identical.

4.2 Euclidean (Baseline)

The Euclidean head applies a 1×1 convolution directly to the backbone features:

$$\hat{\mathbf{y}} = \mathbf{W}_{\text{cls}} \, \mathbf{f} + \mathbf{b}, \quad \mathbf{W}_{\text{cls}} \in \mathbb{R}^{C \times 64}.$$

This is the standard FCN classifier; no geometric embedding is applied.

4.3 Hyperbolic (Ganea / Atigh Baseline)

The hyperbolic head maps backbone features into the Poincaré ball $\mathbb{B}_c^d$ of curvature c using a Möbius transformation, then applies the hyperbolic multinomial logistic regression (MLR) of Ganea et al. [3] extended to the pixel level by Atigh et al. [4]. The decision boundary for class y is a *geodesic hyperplane* in $\mathbb{B}_c^d$.

The curvature c is a hyperparameter optimised during search.

4.4 Pontryagin (This Work)

Motivation. The Poincaré ball has constant negative curvature, which biases the space towards tree-like structures. A Pontryagin space Π_κ is flat but has an *indefinite* metric; it can simultaneously encode positive- and negative-curvature regions through the balance of its positive and negative subspace dimensions.

Definition 1 (Pontryagin space Π_κ)**.** A Pontryagin space of signature κ is a real vector space $\mathbb{R}^{p+\kappa}$ equipped with the indefinite inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle_J = \sum_{i=1}^{p} u_i v_i - \sum_{i=p+1}^{p+\kappa} u_i v_i = \mathbf{u}^\top J \, \mathbf{v}, \quad J = \operatorname{diag}(\underbrace{+1, \ldots, +1}_{p}, \underbrace{-1, \ldots, -1}_{\kappa}).$$

Random feature construction. Given a backbone feature vector $\mathbf{f} \in \mathbb{R}^{64}$ at each pixel, the Pontryagin embedding is built as follows.

Positive component — RFF. Random Fourier Features [1] approximate the RBF kernel

$$K_+(\mathbf{x}, \mathbf{y}) = \exp\left(-\frac{\|\mathbf{x}-\mathbf{y}\|^2}{2\sigma^2}\right).$$

Frequencies $\omega_i \sim \mathcal{N}(0, \sigma^{-2} I)$ and phases $b_i \sim \text{Uniform}[0, 2\pi]$ are sampled once; the feature map is

$$\varphi_+(\mathbf{x}) = \frac{1}{\sqrt{n_{\text{rff}}}} \Big[\cos(\omega_1^\top \mathbf{x} + b_1),\, \sin(\omega_1^\top \mathbf{x} + b_1),\, \ldots\Big] \in \mathbb{R}^{2n_{\text{rff}}}.$$

The bandwidth σ is stored as a learnable parameter $\log \sigma$, which is updated by gradient descent and therefore *removed from the HPO space.*

Negative component — SRF. Spherical Random Features [2] approximate the polynomial kernel

$$K_-(\mathbf{x}, \mathbf{y}) = (\mathbf{x}^\top \mathbf{y})^{d_{\text{poly}}},$$

where d_{poly} is the polynomial degree (*independent of the signature* κ). The j-th feature is

$$\varphi_-(\mathbf{x})_j = \frac{1}{\sqrt{\kappa}} \prod_{\ell=1}^{d_{\text{poly}}} \big(\omega_{j\ell}^\top \mathbf{x}\big), \quad \omega_{j\ell} \sim \text{Uniform}(S^{d-1}),$$

producing $\varphi_-(\mathbf{x}) \in \mathbb{R}^\kappa$.

Concatenation. The Pontryagin embedding is

$$\Phi(\mathbf{x}) = \big[\varphi_+(\mathbf{x}) \,\|\, \varphi_-(\mathbf{x})\big] \in \mathbb{R}^{p+\kappa}, \quad p = 2\, n_{\text{rff}}.$$

The metric matrix $J = \text{diag}(+1^p, -1^\kappa)$ is stored as a fixed buffer.

Remark 1. The signature κ and the polynomial degree d_{poly} are **decoupled hyperparameters**: κ sets how many negative directions exist in J (and therefore how many SRF components are produced), while d_{poly} controls the nonlinearity of the kernel K_-. Both are optimised independently in the HPO search.

J-hyperplane classification. For each class y, the logit is the indefinite inner product with a learnable weight vector $\mathbf{W}_y \in \mathbb{R}^{p+\kappa}$:

$$\zeta_y(\mathbf{z}) = \langle \mathbf{W}_y,\, \mathbf{z} \rangle_J + b_y = \sum_{d=1}^{p+\kappa} J_d\, W_{y,d}\, z_d + b_y.$$

5 Loss Function

The Pontryagin head minimises a composite loss:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{topo}}\, \mathcal{L}_{\text{topo}} + \lambda_{\text{bal}}\, \mathcal{L}_{\text{balance}}.$$

Remark 2 (Cone penalty disabled)**.** An earlier version of the loss included a cone penalty on the classifier weight vectors:

$$\mathcal{L}_{\text{cone-W}} = \frac{1}{C} \sum_{y=1}^{C} \text{ReLU}\big(\varepsilon - \big|\mathbf{W}_y^\top J\, \mathbf{W}_y\big|\big).$$

Empirically, this term suppresses the negative subspace by keeping all weight vectors strictly timelike, which defeats the purpose of the indefinite metric. It has therefore been removed from the active loss ($\lambda_W = 0$); the balance penalty described below serves instead to maintain gradient flow in the negative dimensions.

Cross-entropy.

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{N}\sum_{n=1}^{N} \log \frac{e^{\zeta_{y_n}(\mathbf{z}_n)}}{\sum_c e^{\zeta_c(\mathbf{z}_n)}},$$

weighted by class frequencies ($w_{\mathrm{bg}} = 0.20$, $w_{\mathrm{sc}} = 1.00$).

Topographic penalty $\mathcal{L}_{\mathbf{topo}}$. The `IndefiniteTopographicPenalty` combines three terms on the embedding $\mathbf{z}$:

- *Light-cone (LC):* penalises embeddings near the null cone $\{\mathbf{z} : \langle \mathbf{z}, \mathbf{z}\rangle_J = 0\}$.
- *Causal consistency (CC):* encourages same-class embeddings to share the same causal character (timelike or spacelike).
- *Signature balance (SB):* keeps the norms of the positive and negative subcomponents similar.

Weights $(\lambda_{\mathrm{lc}}, \lambda_{\mathrm{cc}}, \lambda_{\mathrm{sb}}) = (1.0,\ 0.5,\ 0.5)$ are held fixed; λ_{topo} scales the whole term and is searched during HPO.

Balance penalty $\mathcal{L}_{\mathbf{balance}}$.

$$\mathcal{L}_{\mathrm{balance}} = \big(\mathrm{rms}(\mathbf{W}_+) - \mathrm{rms}(\mathbf{W}_-)\big)^2,$$

where $\mathbf{W}_+$ and $\mathbf{W}_-$ are the positive- and negative-subspace columns of the classifier weight matrix, and $\mathrm{rms}(\cdot)$ denotes the root mean square. This penalty encourages the two subspaces to carry equal energy in the weight vectors, ensuring that neither the positive (RFF) nor the negative (SRF) subspace dominates gradient flow during training.

6 Training Protocol

Both experiment tracks (FCN and U-Net) share the same protocol.

Optimiser. Adam with learning rate $\eta = 10^{-3}$ and weight decay 10^{-4} (default values, overridden by the HPO best parameters when available).

Scheduler. Cosine annealing from η to $\eta_{\min} = 10^{-5}$ over $E_{\max} = 60$ epochs.

Early stopping. Training halts if the validation *macro mIoU* (averaged over non-empty classes) does not improve for 15 consecutive epochs. The checkpoint with the highest validation macro mIoU is restored for test evaluation.

Batch size. 8 images per GPU step at 256×256 resolution.

Hardware. Single NVIDIA A100-SXM4 (80 GB) GPU on Google Colab.

7 Hyperparameter Optimisation

Bayesian HPO is carried out with `scikit-optimize`'s `forest_minimize` (Extra Trees surrogate, 30 trials, 10 random starts). Each trial trains for 15 epochs (FCN) or 12 epochs (U-Net). The search spaces differ between the two backbones only in the architecture-depth parameter; all other parameters are shared.

7.1 FCN experiments

Parameter	Range	Prior	Applies to
`lr`	$[10^{-5},\ 10^{-2}]$	log-uniform	all
`weight_decay`	$[10^{-6},\ 10^{-3}]$	log-uniform	all
`backbone_depth`	$\{3, 4, 5\}$	uniform	all
`hyperbolic_c`	$[0.1,\ 2.0]$	log-uniform	hyperbolic
κ	$\{1, \ldots, 64\}$	log-uniform	pontryagin
d_{poly}	$\{1, \ldots, 10\}$	uniform	pontryagin
`rff_multiplier`	$\{1, 2, 3, 4\}$	uniform	pontryagin
λ_{topo}	$[10^{-6},\ 0.2]$	log-uniform	pontryagin
λ_{bal}	$[10^{-3},\ 1.0]$	log-uniform	pontryagin

7.2 U-Net experiments

The U-Net HPO space is identical to the FCN one with a single change: `backbone_depth` is replaced by `unet_depth`, which controls the number of encoder/decoder stages of the U-Net backbone (Section 3.2).

Parameter	Range	Prior	Applies to
`lr`	$[10^{-5},\ 10^{-2}]$	log-uniform	all
`weight_decay`	$[10^{-6},\ 10^{-3}]$	log-uniform	all
`unet_depth`	$\{3, 4, 5\}$	uniform	all
`hyperbolic_c`	$[0.1,\ 2.0]$	log-uniform	hyperbolic
κ	$\{1, \ldots, 64\}$	log-uniform	pontryagin
d_{poly}	$\{1, \ldots, 10\}$	uniform	pontryagin
`rff_multiplier`	$\{1, 2, 3, 4\}$	uniform	pontryagin
λ_{topo}	$[10^{-6},\ 0.2]$	log-uniform	pontryagin
λ_{bal}	$[10^{-3},\ 1.0]$	log-uniform	pontryagin

Design rationale for κ. The signature κ determines how many negative dimensions exist in J and simultaneously the number of SRF output components. A log-uniform prior over $\{1, \ldots, 64\}$ ensures that small values ($\kappa \leq 8$, where the indefinite component is modest) receive the same sampling probability as large values ($\kappa \geq 32$, where the negative subspace is dominant).

Decoupling κ and d_{poly}. In earlier implementations, κ controlled both the number of SRF components and the polynomial degree of K_-, conflating three distinct design choices. The current implementation separates them: $\kappa = n_{\text{srf}}$ sets the signature, while d_{poly} sets the degree of $K_-(\mathbf{x}, \mathbf{y}) = (\mathbf{x}^\top \mathbf{y})^{d_{\text{poly}}}$, searched independently.

Gradient-tuned parameters. The RFF bandwidth σ is *not* part of the HPO space. It is reparametrised as $\sigma = \exp(\log \sigma)$ and updated by gradient descent, with unit-variance frequencies $\omega_i \sim \mathcal{N}(0, I)$ rescaled by $1/\sigma$ at each forward pass.

8 Evaluation Metrics

All metrics are computed on the held-out test split.

Per-class metrics. For each class c: true positives (TP_c), false positives (FP_c), false negatives (FN_c), and true negatives (TN_c); then

$$\mathrm{IoU}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}, \qquad \mathrm{Dice}_c = \frac{2\,\mathrm{TP}_c}{2\,\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}.$$

Precision, recall, and specificity (TNR) are also reported.

Global metrics.

- *Macro mIoU / Dice*: average over non-empty classes.
- *Micro IoU / Dice*: pooled across all pixels.
- *Pixel accuracy*: fraction of correctly labelled pixels.

The monitor criterion for early stopping and HPO is the **validation macro mIoU**.

9 Interpretability: SegScoreCAM

9.1 Motivation

The original ScoreCAM [5] was designed for image *classification*: the score for each feature-map channel is the scalar class logit after masking. Segmentation heads produce spatial output maps $\hat{P} \in \mathbb{R}^{C \times H \times W}$; extracting a single logit discards all spatial information.

9.2 SegScoreCAM

We replace the scalar score with the **mean softmax probability** of the target class over all spatial positions:

$$s_k = \frac{1}{HW} \sum_{i,j} \mathrm{softmax}\big(f_\theta(I \odot \tilde{A}_k)\big)_{c,i,j},$$

where $\tilde{A}_k$ is the k-th feature map normalised to $[0, 1]$ and upsampled to input resolution, and $I \odot \tilde{A}_k$ is the masked input.

The class activation map for class c is then:

$$\mathrm{CAM}_c = \mathrm{ReLU}\left(\sum_k \bar{s}_k \cdot A_k\right), \quad \bar{s}_k = \frac{|s_k|}{\sum_{k'} |s_{k'}|},$$

followed by min-max normalisation to $[0, 1]$.

9.3 Interpretability Metrics

All metrics are computed on the sugarcane class (class 1), where the ground-truth mask $M \in \{0,1\}^{H \times W}$ is available.

Adaptive binarisation (Otsu). Earlier evaluations binarised the CAM at a fixed threshold of 0.5. This is inappropriate when heatmaps have very different dynamic ranges across models. We replace it with **Otsu's method**: the threshold $\tau^* \in [0,1]$ that maximises the inter-class variance of the CAM histogram, computed independently for each image. Let $\hat{M} = \mathbf{1}[\text{CAM} \geq \tau^*]$ be the resulting binary map.

GT-alignment metrics.

Metric	Formula	Interpretation
GT IoU	$\frac{\lvert\hat{M} \cap M\rvert}{\lvert\hat{M} \cup M\rvert}$	overlap with GT ↑
GT Recall	$\frac{\lvert\hat{M} \cap M\rvert}{\lvert M\rvert}$	coverage of GT pixels ↑
GT Precision	$\frac{\lvert\hat{M} \cap M\rvert}{\lvert\hat{M}\rvert}$	fraction of CAM on GT ↑
GT Pearson	$\rho(\text{CAM}_{:},\ M_{:})$	linear correlation (no threshold) ↑
Mean activation	$\frac{1}{HW}\sum_{i,j} \text{CAM}_{i,j}$	diffuseness ↓

Confidence gain. Beyond spatial overlap with the GT mask, we assess whether the highlighted region is *causally sufficient* for the model's prediction by measuring how model confidence changes when only the explanatory region is shown. Given the explanatory image $I_{\text{exp}} = I \odot \text{CAM}$, we compute the per-pixel percentage change in softmax confidence for class c:

$$\Delta_{i,j}^{(c)} = \frac{P_{\text{exp}}(i,j,c) - P_{\text{orig}}(i,j,c)}{P_{\text{orig}}(i,j,c) + \epsilon} \times 100,$$

where P_{orig} and P_{exp} are the softmax outputs on the original and masked images respectively. We then restrict to pixels where the GT label equals c and report the *mean* and *standard deviation* of $\Delta^{(c)}$ over those pixels. A **negative** mean indicates that masking the input by the CAM reduces confidence on the relevant pixels — the model loses information it needs. A value close to zero or positive indicates that the highlighted region is self-sufficient for the prediction.

9.4 Interpretability Results

Table 1 reports all metrics for the sugarcane class. Values marked with ($\star$) are sample estimates from the first recorded test images; full-dataset aggregates require re-running `interpretability_analysis.py` after the HPO re-training with the expanded κ search space.

Several observations stand out.

Otsu binarisation improves all IoU values. Compared to the previous fixed-threshold evaluation (Euclidean GT IoU = 0.287, Hyperbolic = 0.280), the adaptive

Table 1: SegScoreCAM interpretability metrics on the sugarcane class (sample values from available test images, new metric suite with Otsu binarisation). All GT-alignment metrics are higher-is-better; Mean Activation is lower-is-better. Confidence gain is the mean % change in softmax probability on GT-positive pixels when the input is masked by the CAM.

Model	GT IoU ↑	GT Recall ↑	GT Precision ↑	GT Pearson ↑	Mean Act. ↓	Conf. gain (%)
Euclidean	0.658	0.764	0.827	0.836	0.040	-87.0 ± 27.0
Hyperbolic	0.549	0.596	0.874	0.787	0.029	-40.6 ± 31.0
Pontryagin	0.000	0.000	0.000	0.000	0.000	-97.7 ± 1.1

threshold (Otsu $\approx$ 0.29–0.32) yields substantially higher values (Euclidean 0.658, Hyperbolic 0.549), confirming that fixed-threshold comparisons across models with different activation scales are misleading.

Euclidean achieves the best spatial alignment. The Euclidean head leads on GT IoU, Recall, and Pearson correlation, with the Hyperbolic head showing higher GT Precision (0.874 vs 0.827) — suggesting that the hyperbolic embedding produces fewer false-positive activations outside the sugarcane region, at the cost of reduced coverage.

Pontryagin CAM degenerates on the first image. The mean activation of 0.000 and all-zero GT metrics indicate that the SegScoreCAM for the Pontryagin head produces a null heatmap (all feature maps have zero or near-zero softmax contribution) on this image. This is consistent with the earlier finding of high mean activation (0.366) across the dataset: the Pontryagin embedding yields *bimodal* heatmaps — either saturated (diffuse, high mean) or collapsed (null).

Confidence gain is universally negative. All three models show a negative mean confidence change when the input is masked by the sugarcane CAM. This reveals that the models rely on *contextual information* beyond the highlighted region to maintain their predictions. The Hyperbolic model is least affected (-40.6%), suggesting its embedding captures more locally self-sufficient features. The Pontryagin model shows the most severe drop (-97.7%), consistent with its collapsed heatmaps. Importantly, the negative confidence gain is *not* a failure of the CAM method per se — it reflects a property of the underlying model, namely that sugarcane predictions are context-dependent at the scale of the 512×512 image.

10 Discussion

Decoupling κ and d_{poly}. The separation of the Pontryagin signature κ from the SRF polynomial degree d_{poly} is a key design correction. Previously both were set to the same integer, conflating three roles simultaneously: (i) the number of SRF output components, (ii) the number of negative dimensions in J, and (iii) the nonlinearity of K_-. With independent parameters, the HPO can identify whether a modest negative subspace ($\kappa = 2$) paired with a high-degree kernel ($d_{\text{poly}} = 4$) outperforms a high-dimensional subspace ($\kappa = 32$) with a linear kernel ($d_{\text{poly}} = 1$) — something impossible to test under the old parameterisation.

Expanded signature search space. The HPO range for κ has been extended from $\{1, 2, 3, 4\}$ to $\{1, \ldots, 64\}$ with a log-uniform prior. The previous restriction was overly

conservative: with $C_{\text{out}} = 64$ backbone channels and $n_{\text{rff}} = m \cdot 64$, a signature of $\kappa = 4$ produces a Pontryagin space that is strongly dominated by the positive subspace. Larger values of κ allow the indefinite structure to play a more meaningful geometric role. The log-uniform prior ensures that intermediate values (e.g. $\kappa = 8$ or 16) are not underexplored relative to the extremes.

Interpretability findings. Three concrete conclusions emerge from the expanded metric suite:

1. **Fixed-threshold CAM evaluation is unreliable.** The Otsu-adaptive threshold exposes substantially higher GT IoU (0.658 vs 0.287 for Euclidean) compared to the fixed-0.5 baseline, because models with small mean activation require a lower binarisation threshold to capture their concentrated responses.
2. **Hyperbolic embeddings are more locally self-sufficient.** The Hyperbolic head achieves the highest GT Precision (0.874) and the smallest absolute confidence drop (-40.6%), suggesting that the Poincaré-ball embedding concentrates discriminative information more tightly in the highlighted region than the Euclidean or Pontryagin heads.
3. **Pontryagin CAMs collapse under the current HPO regime.** Null heatmaps (all-zero activation) on individual test images, combined with a high mean activation across the dataset, indicate bimodal behaviour: the Pontryagin head alternates between diffuse and collapsed activations depending on the image. This instability is likely driven by the interaction of the cone, topographic, and balance penalties pushing embeddings away from the null cone in a way that disrupts spatial concentration. The expanded κ search and the now-independent d_{poly} are expected to resolve this in subsequent HPO runs.

11 Conclusion

We presented a semantic segmentation framework based on Pontryagin space embeddings, combining Random Fourier Features (RFF) and Spherical Random Features (SRF) with an indefinite J-metric and a J-hyperplane classifier. Three heads — Euclidean, Hyperbolic, and Pontryagin — were trained on a UAV sugarcane segmentation dataset under a controlled protocol (identical backbone, optimiser, and schedule).

Key contributions are:

1. A pixel-level Pontryagin MLR head with composite loss (CE + cone + topographic + balance penalties).
2. Decoupling of the Pontryagin signature κ from the SRF polynomial degree d_{poly}, enabling independent HPO of both.
3. SegScoreCAM: an adaptation of ScoreCAM for segmentation models, with an expanded metric suite comprising Otsu-adaptive binarisation, GT IoU, recall, precision, Pearson correlation, and pixel-level confidence gain on ground-truth class pixels.
4. An interpretability comparison revealing that (a) fixed-threshold CAM evaluation substantially underestimates GT alignment, (b) the hyperbolic head achieves the highest GT precision (0.874) and smallest confidence drop, and (c) the Pontryagin head exhibits bimodal activation behaviour that is expected to stabilise after expanded HPO.

Future work includes a full ablation over $(\kappa, d_{\text{poly}})$ combinations, analysis of the embedding geometry via t-SNE/UMAP in Pontryagin space, investigation of the Pontryagin CAM instability under various composite-loss weight schedules, and extension to multi-class segmentation.

References

[1] A. Rahimi and B. Recht. *Random Features for Large-Scale Kernel Machines*. NeurIPS, 2007.

[2] J. Pennington, F. Yu, and S. Kumar. *Spherical Random Features for Polynomial Kernels*. NeurIPS, 2015.

[3] O.-E. Ganea, G. Bécigneul, and T. Hofmann. *Hyperbolic Neural Networks*. NeurIPS, 2018.

[4] M. G. Atigh, J. Schoep, E. Acar, N. van Noord, and P. Mettes. *Hyperbolic Image Segmentation*. CVPR, 2022.

[5] H. Wang, Z. Wang, M. Du, F. Yang, Z. Zhang, S. Ding, P. Mardziel, and X. Hu. *Score-CAM: Score-Weighted Visual Explanations for Convolutional Neural Networks*. CVPR Workshops, 2020.